

Gopal Kumar

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EDUCATION

University of California, Berkeley

August 2022 - May 2023

Master of Engineering, Industrial Engineering & Operations Research (FinTech)

Relevant Coursework: optimization analytics; risk modeling, simulation & data analysis; applications of machine learning to electronic markets

Indian Institute of Technology, Patna

July 2014 - May 2018

Bachelor of Technology, Computer Science & Engineering

Relevant Coursework: intro to computing; data structures & algorithms; probability theory; software engineering; databases.

TECHNICAL STACK

Languages: Python, R, VBA, AHK, Auto IT, AMPL

Web: Bootstrap, HTML5, JavaScript/jQuery, CSS, Django, Selenium

Databases: MySQL, Oracle

Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, statsmodels

Tools: Excel, EViews, Tableau, Haver Analytics, JPM DataQuery, Refinitiv, Bloomberg, JIRA, Trello, Scrum, Git, Azure, Snowflake

WORK EXPERIENCE

Microsoft

Remote, California

Data Engineering, Sales Analytics, Microsoft 365, Vendor- GJ Consulting

Aug 2023 - Present

- Working in the FastTrack DE team at Microsoft 365 as a vendor employee to drive innovation in Sales products and analytics.

JP Morgan

Mumbai, India

Associate Developer, Economic & Policy Research, US Economics, Corporate & Investment Banking

Jan 2020 – May 2022

- Engineered and implemented Python-based data models, econometric regressions, algorithms, and visualizations as part of data-driven macroeconomic projects, resulting in a 20% increase in project completion efficiency.
- Oversaw the full software development lifecycle (SDLC) for 20+ research projects, culminating in publication-ready notes and interactive real-time dashboards that increased client engagement by 15%.
- Collaborated with media partners like Bloomberg and Yahoo Finance to publish research findings, utilizing Python libraries to create digestible data and statistics, enhancing JP Morgan's brand and attracting 10% more client inquiries.
- Spearheaded the team's efforts in a company-wide DataViz competition, utilizing Python's Matplotlib and Seaborn for data visualization, resulting in a 25% increase in the company's adoption of these techniques winning 3rd position across CIB.
- As a key Python developer, contributed to multiple projects which involved software solutions for the Global Fixed-Income research team, resulting in numerous cutting-edge innovative products which helped us achieve top rankings by institutional investors in 2020 and 2021.

Analyst Developer, Automation & Analytics Strategy, Global Research, Corporate & Investment Banking

Jul 2018 – Dec 2019

- Developed and executed innovative automation and analytics strategies for periodical generation of research reports, web scraping, database setup, and building automation tools, which resulted in a 35% increase in productivity for 1500+ analysts in JP Morgan research.
- Streamlined processes and workflows, saving over 4000+ man-hours annually and increasing team efficiency by 20%.
- As the first full-time Python developer in the team, played a crucial role in defining the cross-functional structures, setting coding standards, and enhancing client-server communications, leading to a 50% increase in the number of projects handled by the team.

KEY PROJECTS

- **“US: Alternative Data-Driven Jobs Tracker”**, 2020-21, JP Morgan. (Tech- Python: scikit-learn)
 - Reimagined the ‘payroll jobs tracking’ by utilizing high-frequency alternative data series (~25 series) to forecast jobs growth (nonfarm payroll).
- **“US: Chase Card Retail Sales Nowcasting”**, 2020-2021, JP Morgan. (Tech- Python: scikit-learn)
 - Led the development of a nowcasting tracker for monthly retail sales using proprietary Chase card consumer spending data.
- **“PDF Tagger and Sentiment Analysis Platform”**, 2019, JP Morgan. (Tech- Django, Python, HTML, CSS, JavaScript)
 - Supervised the design and implementation of a Django-based PDF tagging system, streamlining sentiment analysis and commentary.
- **“JP Morgan Research News Aggregator”**, 2018-2019, JP Morgan. (Tech- Django, Python, HTML, CSS, JavaScript)
 - Directed the creation of a news aggregator system for individual research analysts, integrating RSS feeds, News API, and web scrapers, resulting in a 25% reduction in time spent on information gathering.

NOTABLE PUBLICATIONS

- **“India: A Powell-ian deconstruction of India’s CPI”**, with Sajjid Z Chinoy & Toshi Jain in Bloomberg Quint, Oct 2021, JP Morgan.
 - Coauthored a note discussing a product that assesses inflation trends in India, offering three alternative inflation estimates (‘diffusion index’, ‘trimmed mean index’ & ‘disproportionality index’) and insights into key CPI drivers (~300 time series), garnering 3k+ user engagements.
- **“US: Lessons from a year of alt data watching”**, with Jesse Edgerton & Dan Weitzenfeld, Mar 2021, JP Morgan.
 - Note summarizing utility of alternative data. Concluded high-frequency data provided early signals of dramatic turning points during the pandemic.
- **“Latent dirichlet allocation modeling for CPS patent topic discovery”**, with U.H.Govindarajan & Amy J.C. Trappey in IcoIESE, '18.
 - Implemented LDA model for topic modeling which identified relationships between terms and topics explaining underlying IP dynamics.

AWARDS

- Awarded the **Fung Fellowship** at Fung Institute for Engineering leadership, UC Berkeley (2022), United States. (40,000 USD\$)
- Awarded National Tsing Hua University’s **GEL (Global Engineering Leadership) fellowship** for research (2017), Taiwan. (100,000 NT\$)